LESSON 3.9

Equivalent Ratios

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find and check equivalent ratios using multiplication and division.

WHY THIS MATTERS Chef Montoya is preparing her famous tomato soup for a school banquet. Her original recipe serves 8 people, but tonight she needs to serve 32 guests. She needs your help to scale the recipe without changing the flavor!

REMEMBER

Equivalent Ratios

Proportions & Equivalent Ratios. Formula: a/b = c/d Equivalent ratios are ratios that mean the same thing. You make one by multiplying or dividing BOTH numbers by the same amount.

1. Two ratios are equivalent if their simplified forms or cross products are equal.
2. Find the scale factor between corresponding terms.
3. Multiply or divide to solve for the missing term in the proportion

Words: Equivalent Ratios *(Razones equivalentes)* · Ratio *(Razón)* · Rate *(Tasa)* · Proportion *(Proporción)* · Simplify *(Simplificar)*

1 **Circle the letter of the best answer. Show how you know.**

A printer prints 30 pages every minute. How many pages does it print in 4 minutes?

(A) 7.5 pages
(B) 30 pages
(C) 34 pages
(D) 120 pages

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A faucet fills 3 gallons every 2 minutes. How long does it take to fill 18 gallons?

(A) 6 minutes
(B) 9 minutes
(C) 12 minutes
(D) 36 minutes

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A recipe uses 3 cups of flour for 12 muffins. How much flour is needed for 20 muffins?

(A) 2.4 cups
(B) 4 cups
(C) 5 cups
(D) 6 cups

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Store A sells 5 notebooks for $6.25. Store B sells 8 notebooks for $9.60. Which store has the lower price per notebook?

(A) Store A
(B) They are equal
(C) You cannot tell
(D) Store B

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

A local smoothie shop sells a Berry Blast smoothie. The recipe calls for 2 cups of strawberries for every 3 cups of yogurt. Today they need to make 15 cups of yogurt worth of smoothies for a catering order. What is the scale factor from 3 cups of yogurt to 15 cups?

(A) 3
(B) 5
(C) 12
(D) 45

SHOW YOUR WORK

6 Circle the letter of the best answer. Show how you know.

A local smoothie shop sells a Berry Blast smoothie. The recipe calls for 2 cups of strawberries for every 3 cups of yogurt. Today they need to make 15 cups of yogurt worth of smoothies for a catering order. How many cups of strawberries are needed?

(A) 5
(B) 10
(C) 12
(D) 15

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you find the missing values in Chef Montoya's table? How can you PROVE two of your rows are equivalent ratios?

SENTENCE STARTER I multiplied ___ and ___ by the same number, ___, so the ratio ___ is equivalent to 3:2.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.9

Equivalent Ratios

Set B · Version B · Second form · core practice

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- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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LEARNING TARGET I can find and check equivalent ratios using multiplication and division.

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each recipe ratio to a batch that tastes exactly the same.

_____ 2:3	(A) 8:12
_____ 3:5	(B) 9:15
_____ 4:7	(C) 12:21
_____ 5:6	(D) 10:12

2 **Write the name of the correct group on each line.**

The recipe ratio is 2 cups flour to 5 cups milk (2:5). Sort each batch: does it taste the same (equivalent) or not?

Groups: Equivalent to 2:5 | NOT Equivalent to 2:5

_____ 4:10

_____ 6:15

_____ 8:20

_____ 4:9

_____ 2:7

_____ 5:2

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A local smoothie shop sells a Berry Blast smoothie. The recipe calls for 2 cups of strawberries for every 3 cups of yogurt. Today they need to make 15 cups of yogurt worth of smoothies for a catering order. Why must both ingredients be multiplied by the same factor?

(A) Because 2 and 3 are both small numbers

(B) Because addition would be harder

(C) So the total is a whole number

(D) So the ratio stays 2:3 and the smoothie tastes the same

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Chef Montoya's dressing uses 3 spoons of oil to 4 spoons of vinegar (3:4). Which recipe card shows an EQUIVALENT ratio?

(A) 3:8

(B) 4:3

(C) 6:7

(D) 9:12

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The lemonade station mixes 5 cups of lemon juice to 2 cups of sugar (5:2). Which batch keeps the SAME taste?

(A) 5:4

(B) 7:4

(C) 10:6

(D) 15:6

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — The Scaled-Up Pancake Batter

1. **Set up the ratio** 3 cups flour : 8 cups water = ? : 24 cups water
2. **Find the multiplier** 24 ÷ 8 = 3
3. **Apply to the flour** 3 + 3 = 6
4. **Answer** 6 cups flour : 24 cups water

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you find the missing values in Chef Montoya's table? How can you PROVE two of your rows are equivalent ratios?

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